UNIVERSITY OF SCIENCE & TECHNOLOGY OF HANOI
TRƯỜNG ĐẠI HỌC KHOA HỌC & CÔNG NGHỆ HÀ NỘI

INTERNSHIP TOPIC PROPOSAL

Specialty: ☑ Information and Communication Technology
Academic year: 2023-2026

STUDENT INFORMATION

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Topic: Kitchen Crew: 3D Multiplatform Cooperative Cooking Game

Description:

Kitchen Crew is a teamwork-based game where 1–4 players collaborate in a kitchen to prepare, cook, and serve food orders within a limited time. Players must coordinate their actions to complete different tasks such as collecting ingredients, chopping, cooking, assembling dishes, and delivering meals to customers.

The game features interactive 3D kitchen environments that may include obstacles or moving platforms that challenge players to adapt and communicate effectively. The main objective is to complete as many orders as possible before time runs out while maintaining customer satisfaction. Developed using Unity, the game supports multiplatform gameplay, allowing players on devices such as PC and mobile to play together.

Multiplayer functionality is implemented using Unity Netcode for GameObjects, Unity Lobby, and Unity Relay to support lobby creation, player matchmaking, and cross-platform connectivity.

The difference between this game and Overcooked is that players take the role of both chefs and waiters, requiring them to cook meals and deliver them directly to customers' tables rather than using a delivery counter.

Expected outcomes:

A fully functional 3D cooking game prototype that allows 1–4 players to cooperate in preparing, cooking, and serving food in a shared kitchen environment.

Cross-platform multiplayer support, enabling players on different devices (such as PC and mobile) to connect and play together online.

Interactive 3D kitchen environments with dynamic obstacles and layouts that require players to coordinate and communicate effectively.

Core gameplay systems including: Ingredient preparation/ Cooking mechanics/ Order management/ Scoring and time limits/ Cooperative interactions between players.

User interface and player feedback systems, including timers, order displays, score tracking, and visual/audio feedback during gameplay.

A networking and synchronization system that ensures player actions and game states are synchronized across devices.

Performance optimization for multiple platforms, ensuring the game runs smoothly on both desktop and mobile devices.

A fully playable game demonstrating the core mechanics, cooperative gameplay, and crossplatform multiplayer features.

Used Methods and Techniques:

The game is developed using Unity with the C# programming language. Unity's component-based architecture is used to implement gameplay systems such as player interaction, cooking mechanics, order management, and customer service.

Multiplayer functionality is implemented using Unity Netcode for GameObjects, which synchronizes player actions, kitchen objects, and game states across multiple devices.

Cross-platform multiplayer networking may introduce issues such as network latency, state desynchronization, and connectivity problems between different networks. These issues will be addressed using Unity Netcode for GameObjects, which provides built-in synchronization systems and integrates well with Unity's ecosystem. Compared with alternatives such as Photon Unity Networking and Mirror Networking, Netcode for GameObjects is lightweight and suitable for small cooperative multiplayer projects.

To solve connection issues caused by NAT configurations, the project uses Unity Relay, allowing players to connect without manual port forwarding. Relay is chosen instead of a dedicated server architecture because it simplifies deployment and reduces infrastructure requirements for this project.

To support online multiplayer connections, the system uses Unity Lobby for lobby creation and player session management, allowing users to host or join game rooms.

The gameplay systems are tested through iterative development and playtesting to evaluate player coordination, task management, and the effectiveness of cooperative gameplay mechanics.